A HIGH SPEED/POWER RATIO CONTINUOUS-TIME CMOS CURRENT COMPARATOR

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ABSTRACT

Current comparator is a fundamental component of current-mode circuits. A new high-performance continuous-time CMOS current comparator is proposed in this paper, which comprises one CMOS complementary amplifier, two resistive-load amplifiers and two CMOS inverters. A MOS resistor is used as the CMOS complementary amplifier's negative feedback. Because the voltage swings of the CMOS complementary amplifier are reduced by low input and output resistances, the delay time of the current comparator is shortened. Its power consumption is reduced by the resistive-load amplifiers. Simulation results based on 1.2μm CMOS technology parameters show the speed/power ratio of this new current comparator is better than existing high-performance current comparators. Besides, the new current comparator occupies small area and is process-robust, so it is suitable to high-speed and low-power applications.

1. INTRODUCTION

In recent years current-mode signal processing using CMOS technology has gained great interest[1,2]. Analog CMOS circuits with small area, high speed and low supply voltage have been designed using this approach. Moreover, it is compatible to digital process, so current-mode circuit is considered to be an alternative to voltage-mode circuit for high-speed and low-power applications. As a fundamental component of current-mode analog systems, current comparator can be used in A/D converters, oscillators, current to frequency converters, VLSI neural networks and etc.

The first proposed CMOS continuous-time current comparator[3] is based on high output-resistance cascoded current mirrors, which are connected as a class B stage to amplify small input current difference to large output voltage. Because capacitive load CMOS inverting stages are added in the following to achieve rail-to-rail output voltage, the high output resistance will reduce the speed performance of the current comparator. For high-speed applications, some high-performance continuous-time CMOS current comparators have been proposed in the past few years[4-6]. The current comparator proposed by Traff[4] is shown in Fig. 1(a). I_{in} represents the

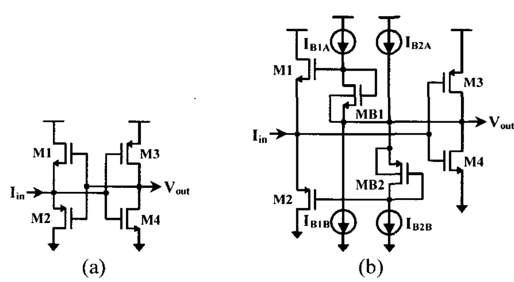

Fig. 1. Conventional CMOS current comparators

difference of the input currents, and V_{out} represents the compared result, which can be

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amplified by following CMOS inverting stages to output a rail-to-rail signal. This current comparator uses a source-follower (M1 and M2) as the input stage and a CMOS inverter (M3 and M4) as the positive feedback, which enables lower input resistance and shorter response time as compared with original current comparator based on current mirrors. But for the dynamic response of small input current, there exists a deadband region temporarily in which the two input transistors (M1 and M2) are both turned off and at this time the input resistance is high. So with the decrease of the input current, the dynamic response time of this current comparator will be dramatically increased. The current comparator in Fig. 1(b) proposed by Tang[5] solves the problem, it changes the biasing scheme of input stage from class B to class AB operation by adding two triode-operation transistors MB1 and MB2, this reduces the deadband region and decreases the response time for small input current. However, this circuit structure is complicated, additional four current sources are needed for biasing circuit, so the power consumption is increased. Besides, the process deviations have great effects on the comparator because its performance is closely related to the values of the four current sources. Furthermore, the bulks of MB1 and MB2 are connected to their sources, not ground and power respectively, so complicated twin-tub CMOS process is needed to implement this circuit. The current comparator proposed by Min[6] comprises three current-source inverting amplifiers and a CMOS inverter. A resistive feedback network is added in the first current-source inverting amplifier to decrease the input resistance. Although this circuit exhibits low response time and good process immunity, its resolution is limited by the bias current. For high-resolution applications, large bias current is needed and power consumption is increased. Moreover, because of the use of current-source amplifiers, its power consumption will not decrease when the input current increases.

In this paper, a new continuous-time current comparator is proposed. Its features include short response delay time, low power consumption, small area and process robustness. Simulation results show its speed/power ratio is better than those of the above current comparators.

2. CIRCUIT DESIGN AND SIMULATION RESULT

The new proposed continuous-time CMOS current comparator is shown in Fig. 2. It

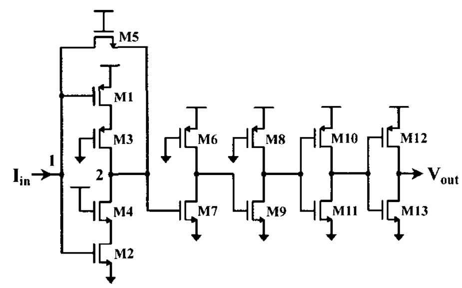

Fig. 2. New CMOS current comparator

comprises one CMOS complementary amplifier (M1-M2), two resistive-load amplifiers (M6-M9) and two CMOS inverters (M10-M13). Because M1 and M2 both work in saturation region, M3 and M4 are used to decrease the working current. The transistor M5 working in linear region acts as the negative feedback resistor of the CMOS complementary amplifier. According to small-signal analysis, when M3 and M4 are neglected, the input and output resistances of the CMOS complementary amplifier with a feedback resistor can be expressed as

$$R_{in} = \frac{R_5 + R_p}{1+\left(g_{m1}+g_{m2}\right)R_p} \qquad (1)$$

$$R_{out} = \frac{R_5 + R_c}{1+\left(g_{m1}+g_{m2}\right)R_c + \left(R_5+R_c\right)/R_p} \qquad (2)$$

where R_p=(r_{ds1}//r_{ds2}), g_{m1} and g_{m2} are the transconductances of M1 and M2 respectively, R_5 is the on-resistance of M5, R_c is output resistance of input current source. Because generally R_5 <<

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R_p, R_5 << R_c and (g_{m1}+ g_{m2}) R_p>>1 exist, it can be concluded that $R_{in}\approx 1/(g_{m1}+g_{m2})$ and $R_{out}\approx 1/(g_{m1}+g_{m2})$. The small input and output resistances can reduce the voltage swings at node 1 and 2, so the response time of the comparator will be greatly decreased. To amplify the small voltage swing at node 2, two resistive-load amplifiers are used to provide additional gain, which can also decrease the power consumption for large input current as compared with current-load amplifiers. The last two CMOS inverters can output a rail-to-rail compared result signal, while the introduced delay time is negligible in practice. This new current comparator has no external bias currents and bias voltages, so the process deviation immunity is enhanced.

To compare the performance of this new current comparator with those of Traff[4], Tang[5], and Min[6], HSPICE simulations of four current comparators are performed using standard 1.2μm CMOS technology parameters with 3V power supply. The transistor dimensions of the proposed circuit are shown in Table 1. The transient responses of the four comparators are simulated for input square-wave currents with amplitudes ranging from ±0.01μA to ±10μA. Fig. 3 shows the transient waveform of the output voltages for the proposed comparator and that of Traff's when the input square-wave current changes between 0.1μA and −0.1μA. It can be seen that the delay time of the proposed comparator is about 8ns, which is ten times lower than that of Traff's. Fig. 4 shows the average delay times of the four comparators as a function of the amplitude of the input square-wave current. For the low input current, the delay time of proposed comparator is much lower than that of Traff's and is comparable with those of Tang's and Min's. When the input current increases, the four delay times decrease with an order of magnitude. When input current is 10μA, the four delay times are almost the same because under this condition the delay time is mainly determined by the propagation delay of the inverting amplifier stages, rather than by the time when enough feedback is applied. Although there is no apparent predominance of the delay time for the proposed comparator over those of Tang's and Min's, its power consumption is lower. Fig. 5 shows the power consumptions of the four comparators as a function of the amplitude of the input square-wave current. Although the power consumption of

Device	M1	M2	M3	M4	M5	M6	M7
W(μm)	7.5	2.4	4	4	2	4	4
L(μm)	1.2	1.2	2	8	8	4	2
Device	M8	M9	M10	M11	M12	M13	
W(μm)	4	4	11.5	4	8.4	3	
L(μm)	4	2	2	2	1.6	1.6	

Table 1. The transistor dimensions of the new circuit

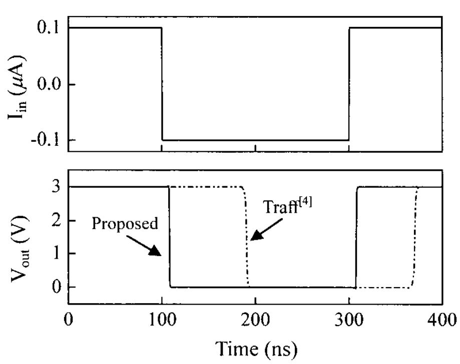

Fig. 3 Transient output voltages against input square-wave current

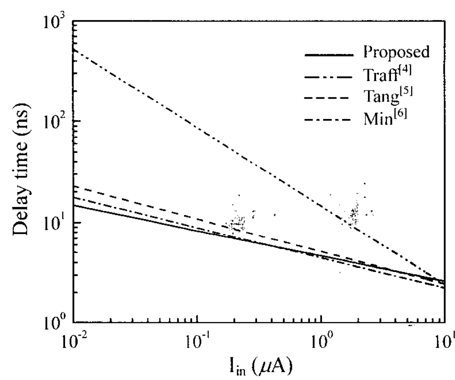

Fig. 4. Average delay times against input current

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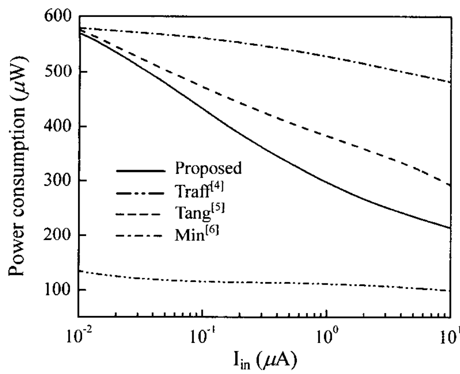

Fig. 5. Power consumptions against input current

Traff's is the lowest, it is achieved with the expense of the large delay time for small input current because its voltage swing is much larger. For the proposed comparator and those of Tang and Min, when the input current is 0.01μA their power consumptions are almost the same. With the increase of the input current, the power consumptions of the proposed comparator and that of Tang's decrease rapidly, while that of Min's almost maintains the same because it uses current-source amplifiers, its bias current does not change when the input current changes. When the input current is 1μA, the power consumption of the proposed comparator and those of Tang's and Min's are 0.3mW, 0.37mW and 0.53mW respectively, their delay times are almost the same, so the speed/power ratio of the proposed comparator is about 127% and 177% over those of Tang's and Min's respectively.

3. CONCLUSIONS

A high-speed and low-power continuous-time CMOS current comparator is designed in this paper, whose delay time is comparable to existing fast current comparators. Because of its lower power consumption, its speed/power ratio is better than other comparators. The simple structure and process robustness also make this circuit suitable to high-speed and low-power applications.

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